

Dionex_COMPASS_September2022

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Daily Set up

Read in and Format the raw data - change wd & file names

```
# SULFATE DATA:
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Sdat <- read.table("Raw Data/COMPASS_Synoptic_CB_MonMon_202209_S04.txt", sep='\t' , header=T, skip=3)
head(Sdat)
```

```
##      X          X.1                X.2 IC.S04 IC.S04.1 IC.S04.2 IC.S04.3 IC.S04.4
## 1 1 Lab Blank      Unknown 4.413    0.5525    6.21    0.4551    2.87
## 2 2 Lab Blank      Unknown 4.417    0.5520    6.21    0.4546    2.87
## 3 3 Lab Blank      Unknown  n.a.      n.a.      n.a.      n.a.      n.a.
## 4 4 Lab Blank      Unknown  n.a.      n.a.      n.a.      n.a.      n.a.
## 5 5 Lab Blank      Unknown 4.423    0.5525    6.20    0.4550    2.87
## 6 6 Standard 1 Calibration Standard 4.427  1.1273    6.27    0.9285    5.72
```

```
## Only keep the columns that we need
```

```
Sdat <- Sdat[,c(2,5)] # dont need this here
head(Sdat)
```

```
##      X.1 IC.S04.1
## 1 Lab Blank 0.5525
## 2 Lab Blank 0.5520
## 3 Lab Blank  n.a.
## 4 Lab Blank  n.a.
## 5 Lab Blank 0.5525
## 6 Standard 1 1.1273
```

```
## Name the columns correctly
```

```
colnames(Sdat) <- c("Sample_ID", "S04_ppm")
Sdat$Sample_ID <- as.factor(Sdat$Sample_ID)
Sdat$S04_ppm <- as.numeric(Sdat$S04_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Sdat <- as.data.frame(Sdat)
head(Sdat)
```

```
##      Sample_ID S04_ppm
## 1 Lab Blank  0.5525
## 2 Lab Blank  0.5520
## 3 Lab Blank      NA
## 4 Lab Blank      NA
## 5 Lab Blank  0.5525
## 6 Standard 1  1.1273
```

```
#Chloride data
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Cldat <- read.table("Raw Data/COMPASS_Synoptic_CB_MonMon_202209_Cl.txt",sep='\t' , header=T, skip=3)
head(Cldat)
```

```
##      X      X.1      X.2 IC.Cl IC.Cl.1 IC.Cl.2 IC.Cl.3 IC.Cl.4
## 1 1 Lab Blank      Unknown 3.293  6.0694  93.08  6.8202  47.25
## 2 2 Lab Blank      Unknown 3.297  6.0646  93.06  6.8149  47.26
## 3 3 Lab Blank      Unknown  n.a.    n.a.    n.a.    n.a.    n.a.
## 4 4 Lab Blank      Unknown 3.290  0.0090 100.00  0.0101  0.07
## 5 5 Lab Blank      Unknown 3.300  6.0747  93.07  6.8262  47.33
## 6 6 Standard 1 Calibration Standard 3.300 12.2365 92.93 13.7503 94.39
```

```
## Only keep the columns that we need
```

```
Cldat <- Cldat[ ,c(2,5)]
head(Cldat)
```

```
##      X.1 IC.Cl.1
## 1 Lab Blank 6.0694
## 2 Lab Blank 6.0646
## 3 Lab Blank  n.a.
## 4 Lab Blank 0.0090
## 5 Lab Blank 6.0747
## 6 Standard 1 12.2365
```

```
## Name the columns correctly
```

```
colnames(Cldat) <- c( "Sample_ID", "Cl_ppm")
Cldat$Sample_ID <- as.factor(Cldat$Sample_ID)
Cldat$Cl_ppm <- as.numeric(Cldat$Cl_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Cldat <- as.data.frame(Cldat)
head(Cldat)
```

```
##      Sample_ID Cl_ppm
## 1 Lab Blank  6.0694
## 2 Lab Blank  6.0646
## 3 Lab Blank      NA
## 4 Lab Blank  0.0090
## 5 Lab Blank  6.0747
## 6 Standard 1 12.2365
```

```
## Bring the data back together:
all_dat <- merge(Sdat, Cldat, by="Sample_ID")
head(all_dat)
```

```
##      Sample_ID   S04_ppm   Cl_ppm
## 1          2568.7298 19169.228
## 2          2568.7298  4914.293
## 3          2568.7298    0.009
## 4          2568.7298  4649.136
## 5          2568.7298     NA
## 6          607.8371 19169.228
```

```
## Remove empty lines
all_dat <- all_dat[!(is.na(all_dat$Sample_ID) | all_dat$Sample_ID==""), ]
head(all_dat)
```

```
##              Sample_ID   S04_ppm   Cl_ppm
## 26      1_202209_MSM_UP_LysA_20cm 185.4786 2155.484
## 27     10_202209_MSM_TR_LysA_20cm 505.3590 3550.881
## 28 10_202209_MSM_TR_LysA_20cm_spk 627.6741 3531.850
## 29      100_202209_MSM_WC_RHZ_6   794.4041 8421.266
## 30     100_202209_MSM_WC_RHZ_6_spk 1072.7880 8331.226
## 31      101_202209_MSM_WC_RHZ_7   815.7621 8675.486
```

```
all_dat <- droplevels(all_dat[!all_dat$Sample_ID == 'Lab Blank',])
head(all_dat)
```

```
##              Sample_ID   S04_ppm   Cl_ppm
## 26      1_202209_MSM_UP_LysA_20cm 185.4786 2155.484
## 27     10_202209_MSM_TR_LysA_20cm 505.3590 3550.881
## 28 10_202209_MSM_TR_LysA_20cm_spk 627.6741 3531.850
## 29      100_202209_MSM_WC_RHZ_6   794.4041 8421.266
## 30     100_202209_MSM_WC_RHZ_6_spk 1072.7880 8331.226
## 31      101_202209_MSM_WC_RHZ_7   815.7621 8675.486
```

Fix Sample IDs

```
##Based on sample list order, 6_202209_MSM_TR_LysC_10cm was probably actually 6_202209_MSM_UP_LysC_10cm
all_dat <- all_dat %>%
  mutate(Sample_ID = gsub("6_202209_MSM_TR_LysC_10cm", "6_202209_MSM_UP_LysC_10cm", Sample_ID))

##Remove samples that were ran several times and had different values
all_dat <- all_dat %>%
  filter(!str_detect(Sample_ID, "29_202209_GWI_UP_LysA_20cm")) %>%
  filter(!str_detect(Sample_ID, "30_202209_GWI_UP_LysA_20cm"))
```

Pull out standards - could do some checks here if we want

```
stds <- all_dat[grep1("Standard", all_dat$Sample_ID),]
head(stds)
```

```
##      Sample_ID S04_ppm Cl_ppm
## 138 Standard 1  1.1273 12.2365
## 139 Standard 1  1.1273 12.3306
## 140 Standard 1  1.1273 12.2419
## 141 Standard 1  1.1273  6.0938
## 142 Standard 1  1.1501 12.2365
## 143 Standard 1  1.1501 12.3306
```

```
stds_chk_S <- stds %>%
  group_by(Sample_ID) %>%
  summarise(mean = mean(S04_ppm), sd = sd(S04_ppm))
stds_chk_S$cv <- (stds_chk_S$sd/stds_chk_S$mean)*100
stds_chk_S$flag <- ifelse(stds_chk_S$cv < 2, 'YES', 'NO, rerun')
head(stds_chk_S)
```

```
## # A tibble: 5 x 5
##   Sample_ID mean    sd    cv flag
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1  0.991  0.258 26.1 NO, rerun
## 2 Standard 2  1.89   0.562 29.8 NO, rerun
## 3 Standard 3  7.70   4.08  53.0 NO, rerun
## 4 Standard 4 16.8    4.79  28.6 NO, rerun
## 5 Standard 5 NA      NA     NA  <NA>
```

```
stds_chk_Cl <- stds %>%
  group_by(Sample_ID) %>%
  summarise(mean = mean(Cl_ppm), sd = sd(Cl_ppm))
stds_chk_Cl$cv <- (stds_chk_Cl$sd/stds_chk_Cl$mean)*100
stds_chk_Cl$flag <- ifelse(stds_chk_Cl$cv < 2, 'YES', 'NO, rerun')
head(stds_chk_Cl)
```

```
## # A tibble: 5 x 5
##   Sample_ID mean    sd    cv flag
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1  10.7  2.76 25.8 NO, rerun
## 2 Standard 2  19.8  5.62 28.4 NO, rerun
## 3 Standard 3  77.8 40.7 52.3 NO, rerun
## 4 Standard 4 168.  47.4 28.3 NO, rerun
## 5 Standard 5  NA   NA     NA  <NA>
```

Calculate mmol/L concentrations

```
#remove standards from sample dataframe
sampledat <- all_dat[!grep1("Standard", all_dat$Sample_ID),]
head(sampledat)
```

```
##           Sample_ID   S04_ppm   Cl_ppm
## 1      1_202209_MSM_UP_LysA_20cm 185.4786 2155.484
## 2      10_202209_MSM_TR_LysA_20cm 505.3590 3550.881
## 3 10_202209_MSM_TR_LysA_20cm_spk 627.6741 3531.850
## 4      100_202209_MSM_WC_RHZ_6    794.4041 8421.266
## 5      100_202209_MSM_WC_RHZ_6_spk 1072.7880 8331.226
## 6      101_202209_MSM_WC_RHZ_7    815.7621 8675.486
```

```
# Constants needed for calculations:
```

```
clmw <- 35.45      #molecular weight of Chloride: 35.45
```

```
smw <- 32.06       #molecular weight of sulfur: 32.06
```

```
# Convert ppm to mmol/L
```

```
sampldat$S04_mM <- (sampldat$S04_ppm / smw)
```

```
sampldat$Cl_mM <- (sampldat$Cl_ppm / clmw)
```

```
# Calculate Salinity
```

```
# calculated using the Knudsen equation
```

```
# Salinity = 0.03 + 1.8050 * Chlorinity
```

```
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
```

```
# =((1.807*Cl_ppm)+0.026)/1000
```

```
sampldat$salinity <- ((1.8070 * sampldat$Cl_ppm) + 0.026) / 1000
```

```
head(sampldat)
```

```
##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1      1_202209_MSM_UP_LysA_20cm 185.4786 2155.484  5.785359  60.80349
## 2      10_202209_MSM_TR_LysA_20cm 505.3590 3550.881 15.762913 100.16590
## 3 10_202209_MSM_TR_LysA_20cm_spk 627.6741 3531.850 19.578107  99.62906
## 4      100_202209_MSM_WC_RHZ_6    794.4041 8421.266 24.778668 237.55333
## 5      100_202209_MSM_WC_RHZ_6_spk 1072.7880 8331.226 33.461884 235.01341
## 6      101_202209_MSM_WC_RHZ_7    815.7621 8675.486 25.444857 244.72456
##      salinity
## 1  3.894985
## 2  6.416469
## 3  6.382079
## 4 15.217253
## 5 15.054550
## 6 15.676629
```

Pull out dups and check with percent difference

```
#Show me the data that we have from the calculations
```

```
head(sampldat)
```

```
##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1      1_202209_MSM_UP_LysA_20cm 185.4786 2155.484  5.785359  60.80349
## 2      10_202209_MSM_TR_LysA_20cm 505.3590 3550.881 15.762913 100.16590
```

```
## 3 10_202209_MSM_TR_LysA_20cm_spk 627.6741 3531.850 19.578107 99.62906
## 4      100_202209_MSM_WC_RHZ_6 794.4041 8421.266 24.778668 237.55333
## 5      100_202209_MSM_WC_RHZ_6_spk 1072.7880 8331.226 33.461884 235.01341
## 6      101_202209_MSM_WC_RHZ_7 815.7621 8675.486 25.444857 244.72456
##      salinity
## 1 3.894985
## 2 6.416469
## 3 6.382079
## 4 15.217253
## 5 15.054550
## 6 15.676629
```

```
#pull out any rows that have "dup" in the SampleID column
dups <- sampledats %>%
  filter(str_detect(Sample_ID, "dup")) #have to change this to match data
head(dups)
```

```
##              Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 109_202209_GWI_TR_RHZ_SF_7_dup 933.3051 6544.394 29.1112009 184.6091312
## 2 19_202209_MSM_WC_SipA_45cm_dup 51.6200 8507.394 1.6101061 239.9828999
## 3 39_202209_GWI_TR_LysA_45cm_dup 701.8766 8857.495 21.8925951 249.8588124
## 4 49_202209_GWI_WC_SipB_10cm_dup 1907.2701 13781.568 59.4906457 388.7607419
## 5 59_202209_GCrew_TR_LysA_20cm_dup 10.1312 3.942 0.3160075 0.1111989
## 6 69_202209_GCrew_WC_SipA_45cm_dup 51.9954 5737.635 1.6218153 161.8514781
##      salinity
## 1 11.825745416
## 2 15.372886597
## 3 16.005519284
## 4 24.903319918
## 5 0.007149194
## 6 10.367932264
```

```
#remove these from sample dataframe in a new dataframe
sampledat2 <- sampledats %>%
  filter(!str_detect(Sample_ID, "dup")) %>%
  filter(!str_detect(Sample_ID, "spk"))
head(sampledat2)
```

```
##              Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM  salinity
## 1 1_202209_MSM_UP_LysA_20cm 185.4786 2155.484 5.785359 60.80349 3.894985
## 2 10_202209_MSM_TR_LysA_20cm 505.3590 3550.881 15.762913 100.16590 6.416469
## 3 100_202209_MSM_WC_RHZ_6 794.4041 8421.266 24.778668 237.55333 15.217253
## 4 101_202209_MSM_WC_RHZ_7 815.7621 8675.486 25.444857 244.72456 15.676629
## 5 102_202209_MSM_WC_RHZ_8 872.0692 8465.177 27.201160 238.79201 15.296600
## 6 103_202209_GWI_TR_RHZ_SF_1 951.8538 6184.023 29.689763 174.44353 11.174556
```

```
#remove the dup from these IDs so we will have duplicate sample names
dups$Sample_ID <- gsub("_dup", "", as.character(dups$Sample_ID))
dups <- dups[, -c(2,3)]
colnames(dups) <- c('Sample_ID', 'S04_mM_dup', 'Cl_mM_dup', 'salinity_dup')
head(dups)
```

```
##           Sample_ID S04_mM_dup  Cl_mM_dup salinity_dup
## 1  109_202209_GWI_TR_RHZ_SF_7 29.1112009 184.6091312 11.825745416
## 2   19_202209_MSM_WC_SipA_45cm 1.6101061 239.9828999 15.372886597
## 3   39_202209_GWI_TR_LysA_45cm 21.8925951 249.8588124 16.005519284
## 4   49_202209_GWI_WC_SipB_10cm 59.4906457 388.7607419 24.903319918
## 5  59_202209_GCrew_TR_LysA_20cm 0.3160075  0.1111989  0.007149194
## 6  69_202209_GCrew_WC_SipA_45cm 1.6218153 161.8514781 10.367932264
```

```
#put it back together with the old data set and look for duplicates
QAdups <- merge(sampled2, dups)
head(QAdups)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1  109_202209_GWI_TR_RHZ_SF_7 927.5268 6420.4224 28.9309669 181.1120564
## 2   19_202209_MSM_WC_SipA_45cm 52.8767 8548.8314 1.6493044 241.1518025
## 3   39_202209_GWI_TR_LysA_45cm 689.8972 8687.1963 21.5189395 245.0549027
## 4   49_202209_GWI_WC_SipB_10cm 1925.9192 13905.9219 60.0723394 392.2686008
## 5  59_202209_GCrew_TR_LysA_20cm 10.4618  4.9391  0.3263194  0.1393258
## 6  69_202209_GCrew_WC_SipA_45cm 50.8383 5615.6024 1.5857236 158.4090945
##           salinity S04_mM_dup  Cl_mM_dup salinity_dup
## 1 11.601729277 29.1112009 184.6091312 11.825745416
## 2 15.447764340 1.6101061 239.9828999 15.372886597
## 3 15.697789714 21.8925951 249.8588124 16.005519284
## 4 25.128026873 59.4906457 388.7607419 24.903319918
## 5  0.008950954 0.3160075  0.1111989  0.007149194
## 6 10.147419537 1.6218153 161.8514781 10.367932264
```

```
QAdups$S04_dups_chk <- ((abs(QAdups$S04_mM-QAdups$S04_mM_dup))/((QAdups$S04_mM+QAdups$S04_mM_dup)/2))*100
QAdups$S04_dups_flag <- ifelse(QAdups$S04_dups_chk <10, 'YES', 'NO, rerun')
```

```
QAdups$Cl_dups_chk <- ((abs(QAdups$Cl_mM-QAdups$Cl_mM_dup))/((QAdups$Cl_mM+QAdups$Cl_mM_dup)/2))*100
QAdups$Cl_dups_flag <- ifelse(QAdups$Cl_dups_chk <10, 'YES', 'NO, rerun')
```

```
head(QAdups)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1  109_202209_GWI_TR_RHZ_SF_7 927.5268 6420.4224 28.9309669 181.1120564
## 2   19_202209_MSM_WC_SipA_45cm 52.8767 8548.8314 1.6493044 241.1518025
## 3   39_202209_GWI_TR_LysA_45cm 689.8972 8687.1963 21.5189395 245.0549027
## 4   49_202209_GWI_WC_SipB_10cm 1925.9192 13905.9219 60.0723394 392.2686008
## 5  59_202209_GCrew_TR_LysA_20cm 10.4618  4.9391  0.3263194  0.1393258
## 6  69_202209_GCrew_WC_SipA_45cm 50.8383 5615.6024 1.5857236 158.4090945
##           salinity S04_mM_dup  Cl_mM_dup salinity_dup S04_dups_chk S04_dups_flag
## 1 11.601729277 29.1112009 184.6091312 11.825745416  0.6210448      YES
## 2 15.447764340 1.6101061 239.9828999 15.372886597  2.4052434      YES
## 3 15.697789714 21.8925951 249.8588124 16.005519284  1.7214579      YES
## 4 25.128026873 59.4906457 388.7607419 24.903319918  0.9730331      YES
## 5  0.008950954 0.3160075  0.1111989  0.007149194  3.2107998      YES
## 6 10.147419537 1.6218153 161.8514781 10.367932264  2.2504296      YES
##           Cl_dups_chk Cl_dups_flag
## 1  1.9124267      YES
## 2  0.4858941      YES
## 3  1.9413120      YES
```

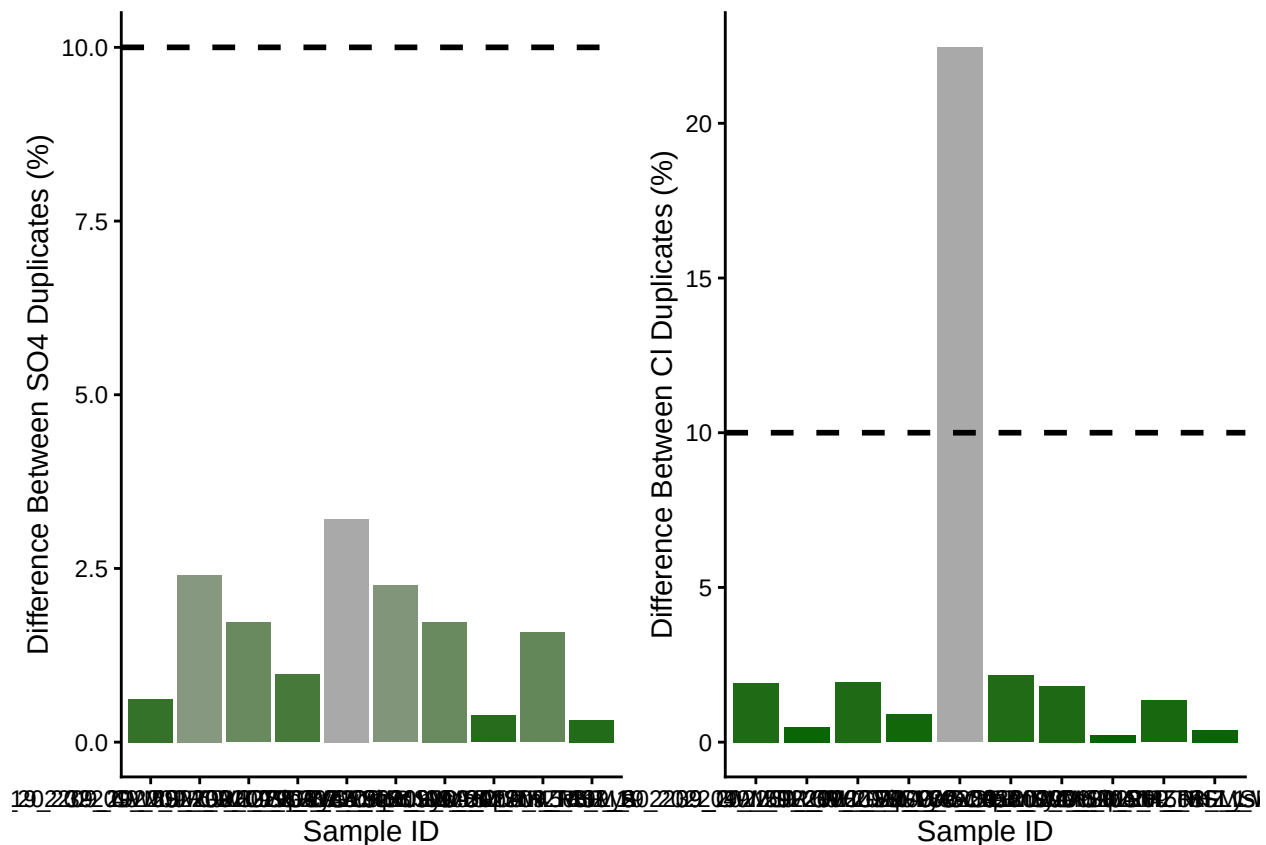
```
## 4 0.8982656 YES
## 5 22.4544257 NO, rerun
## 6 2.1497393 YES
```

```
#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_chk, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between Cl Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar,ncol=2, nrow=1)
```




```

#check for percent of no, reruns to see if it would warrant reruns
Perc_dups <- QAdups %>%
  group_by(S04_dups_flag) %>%
  summarise(S_no_rows = length(S04_dups_flag))
Perc_dups1 <- QAdups %>%
  group_by(Cl_dups_flag) %>%
  summarise(Cl_no_rows = length(Cl_dups_flag))
colnames(Perc_dups) <- c("Flag", "S_no_rows")
colnames(Perc_dups1) <- c("Flag", "Cl_no_rows")
Perc_dups2 <- cbind(Perc_dups, Perc_dups1)

Perc_dups2$Total <- length(QAdups$S04_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$Cl_Percent <- (Perc_dups2$Cl_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)

```

```

##   Flag S_no_rows      Flag Cl_no_rows Total S_Percent Cl_Percent
## 1  YES         10 NO, rerun         1     10         100         10
## 2  YES         10     YES         9     10         100         90

```

Pull out dups and check with cv

```

#the cv
# calculate the sd between the two duplicates then divide by the mean and multiply by 100

df2 <- as.data.frame(QAdups$S04_mM)
df2$dups <- QAdups$S04_mM_dup

df2$sds <- apply(df2,1,sd)

QAdups$S04_dups_cv <- (df2$sds)/((QAdups$S04_mM+QAdups$S04_mM_dup)/2) * 100
QAdups$S04_dups_cv_flag <- ifelse(QAdups$S04_dups_cv <11, 'YES', 'NO, rerun')

head(QAdups)

```

```

##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1  109_202209_GWI_TR_RHZ_SF_7  927.5268  6420.4224  28.9309669  181.1120564
## 2   19_202209_MSM_WC_SipA_45cm   52.8767  8548.8314   1.6493044  241.1518025
## 3   39_202209_GWI_TR_LysA_45cm  689.8972  8687.1963  21.5189395  245.0549027
## 4   49_202209_GWI_WC_SipB_10cm 1925.9192 13905.9219  60.0723394  392.2686008
## 5  59_202209_GCrew_TR_LysA_20cm   10.4618    4.9391   0.3263194   0.1393258
## 6  69_202209_GCrew_WC_SipA_45cm   50.8383  5615.6024   1.5857236  158.4090945
##           salinity S04_mM_dup  Cl_mM_dup salinity_dup S04_dups_chk S04_dups_flag
## 1 11.601729277 29.1112009 184.6091312 11.825745416   0.6210448         YES
## 2 15.447764340  1.6101061 239.9828999 15.372886597   2.4052434         YES
## 3 15.697789714 21.8925951 249.8588124 16.005519284   1.7214579         YES
## 4 25.128026873 59.4906457 388.7607419 24.903319918   0.9730331         YES
## 5  0.008950954  0.3160075  0.1111989  0.007149194   3.2107998         YES
## 6 10.147419537  1.6218153 161.8514781 10.367932264   2.2504296         YES
##   Cl_dups_chk Cl_dups_flag S04_dups_cv S04_dups_cv_flag
## 1   1.9124267         YES   0.4391450         YES

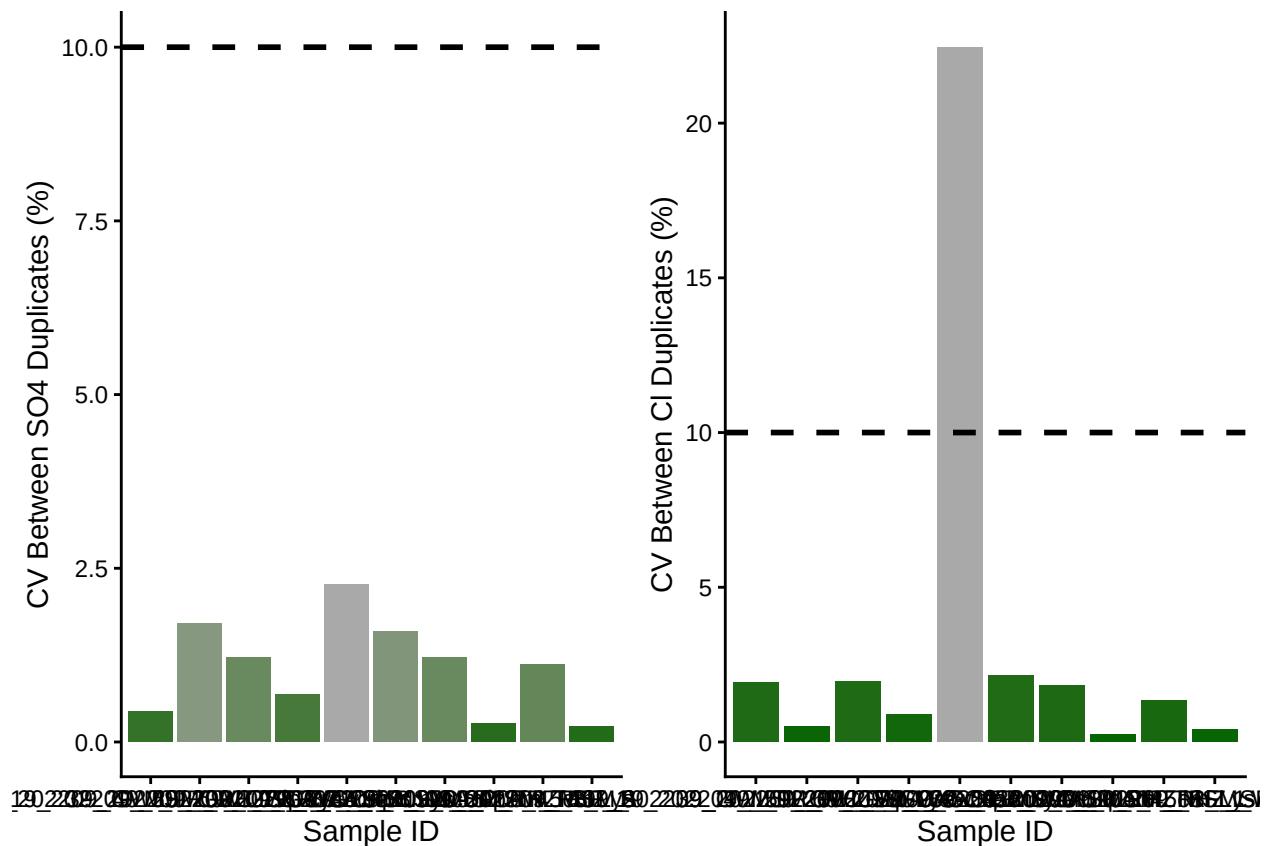
```

## 2	0.4858941	YES	1.7007639	YES
## 3	1.9413120	YES	1.2172546	YES
## 4	0.8982656	YES	0.6880383	YES
## 5	22.4544257	NO, rerun	2.2703783	YES
## 6	2.1497393	YES	1.5912940	YES

```
#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_cv, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="CV Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
    color = "black", size=1)

Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = C1_dups_chk, fill=C1_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="CV Between C1 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
    color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar,ncol=2, nrow=1)
```



```

#check for percent of no, reruns to see if it would warrant reruns
Perc_dups <- QAdups %>%
  group_by(S04_dups_flag) %>%
  summarise(S_no_rows = length(S04_dups_flag))
Perc_dups1 <- QAdups %>%
  group_by(Cl_dups_flag) %>%
  summarise(Cl_no_rows = length(Cl_dups_flag))
colnames(Perc_dups) <- c("Flag", "S_no_rows")
colnames(Perc_dups1) <- c("Flag", "Cl_no_rows")
Perc_dups2 <- cbind(Perc_dups, Perc_dups1)

Perc_dups2$Total <- length(QAdups$S04_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$Cl_Percent <- (Perc_dups2$Cl_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)

```

```

##   Flag S_no_rows      Flag Cl_no_rows Total S_Percent Cl_Percent
## 1  YES         10 NO, rerun         1     10         100         10
## 2  YES         10     YES         9     10         100         90

```

Pull out spikes and check

```

#Show me the data that we have from the calculations
head(sampledat)

```

```

##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1  1_202209_MSM_UP_LysA_20cm 185.4786 2155.484  5.785359 60.80349
## 2  10_202209_MSM_TR_LysA_20cm 505.3590 3550.881 15.762913 100.16590
## 3 10_202209_MSM_TR_LysA_20cm_spk 627.6741 3531.850 19.578107 99.62906
## 4    100_202209_MSM_WC_RHZ_6  794.4041 8421.266 24.778668 237.55333
## 5   100_202209_MSM_WC_RHZ_6_spk 1072.7880 8331.226 33.461884 235.01341
## 6   101_202209_MSM_WC_RHZ_7   815.7621 8675.486 25.444857 244.72456
##   salinity
## 1  3.894985
## 2  6.416469
## 3  6.382079
## 4 15.217253
## 5 15.054550
## 6 15.676629

```

```

#pull out any rows that have "spk" in the SampleID column
spks <- sampledat %>%
  filter(str_detect(Sample_ID, "spk")) #have to change this to match data
head(spks)

```

```

##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1 10_202209_MSM_TR_LysA_20cm_spk 627.6741 3531.850 19.57811 99.62906
## 2   100_202209_MSM_WC_RHZ_6_spk 1072.7880 8331.226 33.46188 235.01341
## 3 110_202209_GWI_TR_RHZ_SF_8_spk  793.6349 4702.940 24.75468 132.66403
## 4  20_202209_MSM_WC_SipB_10cm_spk 1122.1770 9663.288 35.00240 272.58922

```

```
## 5 40_202209_GWI_TR_LysB_10cm_spk 1990.9739 12742.654 62.10149 359.45428
## 6 50_202209_GWI_WC_SipB_20cm_spk 2265.0899 12474.505 70.65159 351.89014
##      salinity
## 1  6.382079
## 2 15.054550
## 3  8.498238
## 4 17.461587
## 5 23.026002
## 6 22.541457
```

```
#remove the dup from these IDs so we will have duplicate sample names
spks$Sample_ID<-gsub("_spk","",as.character(spks$Sample_ID))
spks <- spks[ ,-c(2,3, 5,6)]
colnames(spks) <- c('Sample_ID', 'S04_mM_spk')
head(spks)
```

```
##              Sample_ID S04_mM_spk
## 1 10_202209_MSM_TR_LysA_20cm  19.57811
## 2   100_202209_MSM_WC_RHZ_6  33.46188
## 3 110_202209_GWI_TR_RHZ_SF_8  24.75468
## 4 20_202209_MSM_WC_SipB_10cm  35.00240
## 5 40_202209_GWI_TR_LysB_10cm  62.10149
## 6 50_202209_GWI_WC_SipB_20cm  70.65159
```

```
#put it back together with the old data set and look for duplicates
QAspks <- merge(sampldat, spks)
head(QAspks)
```

```
##              Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM  salinity
## 1 10_202209_MSM_TR_LysA_20cm 505.3590 3550.881 15.76291 100.1659 6.416469
## 2   100_202209_MSM_WC_RHZ_6 794.4041 8421.266 24.77867 237.5533 15.217253
## 3 110_202209_GWI_TR_RHZ_SF_8 962.0158 6500.256 30.00673 183.3641 11.745989
## 4 20_202209_MSM_WC_SipB_10cm 860.1581 9655.185 26.82964 272.3606 17.446945
## 5 40_202209_GWI_TR_LysB_10cm 1698.7295 12719.267 52.98595 358.7946 22.983742
## 6 50_202209_GWI_WC_SipB_20cm 1676.8623 12437.680 52.30388 350.8513 22.474914
##      S04_mM_spk
## 1  19.57811
## 2  33.46188
## 3  24.75468
## 4  35.00240
## 5  62.10149
## 6  70.65159
```

```
#now we need to calculate the spike concentration and calculate the spike recovery
spkconc <- (250/smw)      # in mM
spkvol <- 10              # in uL
spkvol <- spkvol/1000000
#spike for these samples was 10uL of the 250mM standard
QAspks$S04_spk_Conc <- (spkconc)*spkvol      # mmoles of S04
head(QAspks)
```

```
##              Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM  salinity
```

```
## 1 10_202209_MSM_TR_LysA_20cm 505.3590 3550.881 15.76291 100.1659 6.416469
## 2 100_202209_MSM_WC_RHZ_6 794.4041 8421.266 24.77867 237.5533 15.217253
## 3 110_202209_GWI_TR_RHZ_SF_8 962.0158 6500.256 30.00673 183.3641 11.745989
## 4 20_202209_MSM_WC_SipB_10cm 860.1581 9655.185 26.82964 272.3606 17.446945
## 5 40_202209_GWI_TR_LysB_10cm 1698.7295 12719.267 52.98595 358.7946 22.983742
## 6 50_202209_GWI_WC_SipB_20cm 1676.8623 12437.680 52.30388 350.8513 22.474914
## S04_mM_spk S04_spk_Conc
## 1 19.57811 7.797879e-05
## 2 33.46188 7.797879e-05
## 3 24.75468 7.797879e-05
## 4 35.00240 7.797879e-05
## 5 62.10149 7.797879e-05
## 6 70.65159 7.797879e-05
```

```
#need to determine dilution factors and initial amount of sample added
#if your samples are all the same dilution just use the first line
#for Steph / COMPASS this depends on the site so...
```

```
QAspks$Dilution <- 1
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_UP"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_TR"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_WC"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_UP"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_TR"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_WC"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_UP"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_TR"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_WC"), 200, QAspks$Dilution)
#QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_UP"), 100, QAspks$Dilution)
#QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_TR"), 100, QAspks$Dilution)
#QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_WC"), 100, QAspks$Dilution)

#Set Sample volumes in uL
QAspks$SampleVol <- 1
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_UP"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_TR"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_WC"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_UP"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_TR"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_WC"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_UP"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_TR"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_WC"), 1462, QAspks$SampleVol)
#QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_UP"), 100, QAspks$SampleVol)
#QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_TR"), 100, QAspks$SampleVol)
#QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_WC"), 100, QAspks$SampleVol)
```

```
#change sample volume to L
QAspks$SampleVol <- QAspks$SampleVol/1000000
head(QAspks)
```

```
## Sample_ID S04_ppm Cl_ppm S04_mM Cl_mM salinity
## 1 10_202209_MSM_TR_LysA_20cm 505.3590 3550.881 15.76291 100.1659 6.416469
## 2 100_202209_MSM_WC_RHZ_6 794.4041 8421.266 24.77867 237.5533 15.217253
## 3 110_202209_GWI_TR_RHZ_SF_8 962.0158 6500.256 30.00673 183.3641 11.745989
```

```
## 4 20_202209_MSM_WC_SipB_10cm 860.1581 9655.185 26.82964 272.3606 17.446945
## 5 40_202209_GWI_TR_LysB_10cm 1698.7295 12719.267 52.98595 358.7946 22.983742
## 6 50_202209_GWI_WC_SipB_20cm 1676.8623 12437.680 52.30388 350.8513 22.474914
##   S04_mM_spk S04_spk_Conc Dilution SampleVol
## 1   19.57811 7.797879e-05      50 0.001501
## 2   33.46188 7.797879e-05     100 0.001475
## 3   24.75468 7.797879e-05     100 0.001475
## 4   35.00240 7.797879e-05     100 0.001475
## 5   62.10149 7.797879e-05     100 0.001475
## 6   70.65159 7.797879e-05     200 0.001462
```

#gives us the total SO4 in the sample in mmoles

```
QAspks$S04_Total_unspkd <- (QAspks$S04_mM/QAspks$Dilution)*(QAspks$SampleVol)
```

##total SO4 in spiked sample in mmoles

```
QAspks$S04_Total_spkd <- (QAspks$S04_mM_spk/QAspks$Dilution)*(QAspks$SampleVol+spkvol)
```

```
QAspks$S04_expctd_spkd <- (QAspks$S04_Total_unspkd + QAspks$S04_spk_Conc)
```

```
QAspks$spk_recovery <- (QAspks$S04_Total_spkd/QAspks$S04_expctd_spkd)*100
```

```
QAspks$S04_spks_flag <- ifelse(QAspks$spk_recovery <=120 & QAspks$spk_recovery >=80 , 'YES', 'NO, rerun
```

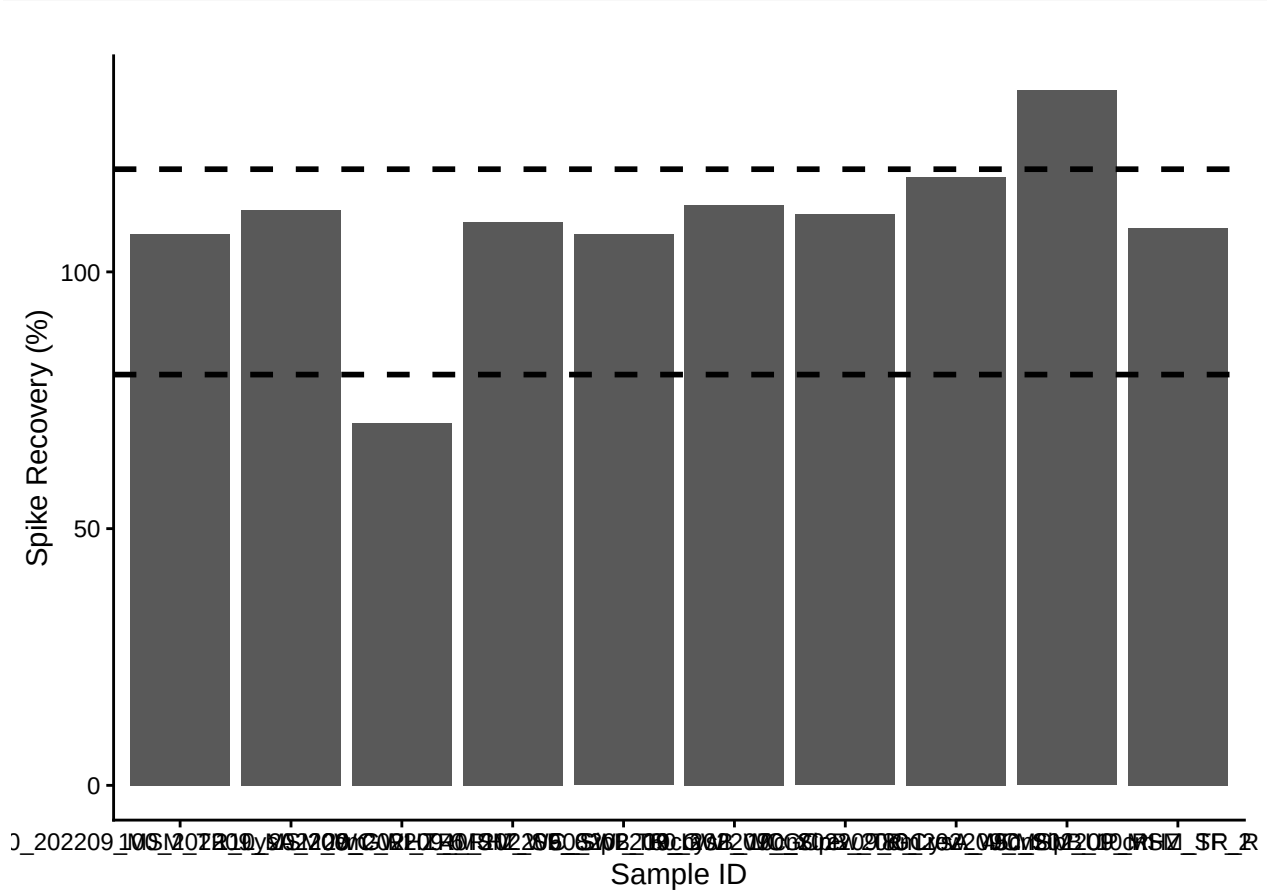
```
head(QAspks)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM  salinity
## 1 10_202209_MSM_TR_LysA_20cm 505.3590 3550.881 15.76291 100.1659 6.416469
## 2 100_202209_MSM_WC_RHZ_6 794.4041 8421.266 24.77867 237.5533 15.217253
## 3 110_202209_GWI_TR_RHZ_SF_8 962.0158 6500.256 30.00673 183.3641 11.745989
## 4 20_202209_MSM_WC_SipB_10cm 860.1581 9655.185 26.82964 272.3606 17.446945
## 5 40_202209_GWI_TR_LysB_10cm 1698.7295 12719.267 52.98595 358.7946 22.983742
## 6 50_202209_GWI_WC_SipB_20cm 1676.8623 12437.680 52.30388 350.8513 22.474914
##   S04_mM_spk S04_spk_Conc Dilution SampleVol S04_Total_unspkd S04_Total_spkd
## 1   19.57811 7.797879e-05      50 0.001501    0.0004732027    0.0005916504
## 2   33.46188 7.797879e-05     100 0.001475    0.0003654854    0.0004969090
## 3   24.75468 7.797879e-05     100 0.001475    0.0004425993    0.0003676069
## 4   35.00240 7.797879e-05     100 0.001475    0.0003957371    0.0005197857
## 5   62.10149 7.797879e-05     100 0.001475    0.0007815427    0.0009222072
## 6   70.65159 7.797879e-05     200 0.001462    0.0003823413    0.0005199957
##   S04_expctd_spkd spk_recovery S04_spks_flag
## 1    0.0005511814    107.34222         YES
## 2    0.0004434641    112.05167         YES
## 3    0.0005205781     70.61514      NO, rerun
## 4    0.0004737159    109.72519         YES
## 5    0.0008595215    107.29309         YES
## 6    0.0004603201    112.96392         YES
```

#plot spk recoveries output as a bar graph to easily check - want any over 10% to be red need to work on

```
spksbar <- ggplot(data = QAspks, aes(x = Sample_ID, y = spk_recovery)) +
  geom_bar(stat = 'identity') +
  scale_fill_manual(values=c("lightblue4")) +
  theme_classic() + labs(x= "Sample ID", y="Spike Recovery (%)") +
  theme(legend.position="none") +
  geom_hline(yintercept=80, linetype="dashed", color = "black", size=1) +
  geom_hline(yintercept=120, linetype="dashed", color = "black", size=1)
```

spksbar



```
#check for percent of no, reruns to see if it would warrant reruns
Perc_spks <- QAspks %>%
  group_by(S04_spks_flag) %>%
  summarise(no_rows = length(S04_spks_flag))
Perc_spks$Total <- length(QAspks$S04_spks_flag)
Perc_spks$Percent <- (Perc_spks$no_rows / Perc_spks$Total)*100
head(Perc_spks)
```

```
## # A tibble: 2 x 4
##   S04_spks_flag no_rows Total Percent
##   <chr>         <int> <int>   <dbl>
## 1 NO, rerun           2    10     20
## 2 YES                 8    10     80
```

Make final dataframe with IDs

```
#for steph <- pull out identifiers of the sample names
#pull the sample ID and separate it by the underscores
IDs <- data.frame(do.call('rbind', strsplit(as.character(sampledat2$Sample_ID), '_', fixed=TRUE)))
```

```
## Warning in rbind(c("1", "202209", "MSM", "UP", "LysA", "20cm"), c("10", :
## number of columns of result is not a multiple of vector length (arg 1)
```

```
colnames(IDs) <- c("Analysis_No" , "Date","Site", "Zone", "Replicate", "Depth", "Tree")
head(IDs)
```

```
##   Analysis_No   Date Site Zone Replicate Depth Tree
## 1           1 202209  MSM  UP        LysA  20cm   1
## 2          10 202209  MSM  TR        LysA  20cm  10
## 3         100 202209  MSM  WC         RHZ    6  100
## 4         101 202209  MSM  WC         RHZ    7  101
## 5         102 202209  MSM  WC         RHZ    8  102
## 6         103 202209  GWI  TR         RHZ    SF   1
```

```
#rejoin them to the dataframe
alldat <- cbind(IDs, sampled2)
head(alldat)
```

```
##   Analysis_No   Date Site Zone Replicate Depth Tree           Sample_ID
## 1           1 202209  MSM  UP        LysA  20cm   1 1_202209_MSM_UP_LysA_20cm
## 2          10 202209  MSM  TR        LysA  20cm  10 10_202209_MSM_TR_LysA_20cm
## 3         100 202209  MSM  WC         RHZ    6  100 100_202209_MSM_WC_RHZ_6
## 4         101 202209  MSM  WC         RHZ    7  101 101_202209_MSM_WC_RHZ_7
## 5         102 202209  MSM  WC         RHZ    8  102 102_202209_MSM_WC_RHZ_8
## 6         103 202209  GWI  TR         RHZ    SF   1 103_202209_GWI_TR_RHZ_SF_1
##   S04_ppm  Cl_ppm   S04_mM   Cl_mM  salinity
## 1 185.4786 2155.484  5.785359 60.80349 3.894985
## 2 505.3590 3550.881 15.762913 100.16590 6.416469
## 3 794.4041 8421.266 24.778668 237.55333 15.217253
## 4 815.7621 8675.486 25.444857 244.72456 15.676629
## 5 872.0692 8465.177 27.201160 238.79201 15.296600
## 6 951.8538 6184.023 29.689763 174.44353 11.174556
```

Make final dataframe with IDs

```
write.csv(alldat, file="Processed Data/COMPASS_SynopticCB_PW_Processed_Cl_S04_202209.csv") #Change fil
```

END